

Agentic Healthcare Maps - Challenge 3

An End-to-End Autonomous Pipeline for Healthcare Facility Intelligence

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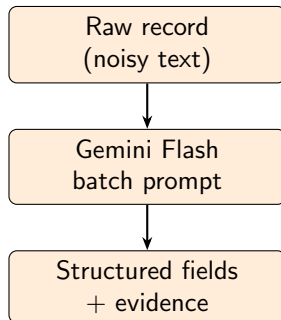
Stage 1 — LLM Extraction with Evidence

Input: Noisy Excel rows

- Inconsistent field names
- Missing values, contradictory columns
- Free-text mixed with structured data

Output: Structured `.jsonl`

- `facility_type`, `beds_count`
- `has_icu`, `has_emergency_trauma`
- `evidence_sentences[]` — verbatim source quotes



Stage 2 — Hybrid Trust Scoring & Self-Correction

Rule score (50%):

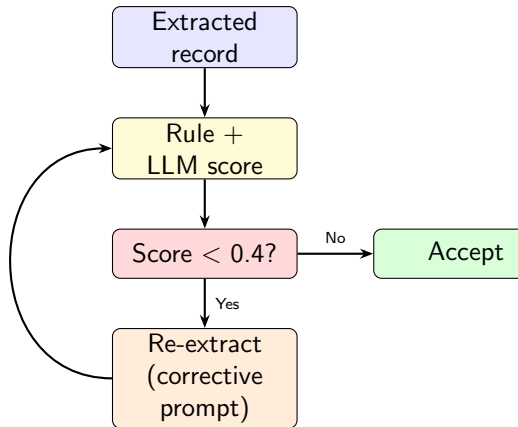
- Field completeness
- Contradiction checks (e.g. PHC + ICU)
- Geocode resolution quality

LLM score (50%):

- Gemini auditor prompt
- Returns `consistency_score` + contradictions list

Self-correction loop:

$\text{Score} < 0.4 \Rightarrow$ re-extract with corrective prompt



Stages 3 & 4 — Geocoding and Interactive Map

Geocoding (Nominatim):

4-level address fallback:

- 1 Full address
- 2 City + State
- 3 State centroid
- 4 Skip (mark unresolved)

Persistent JSON cache avoids repeat requests.

Interactive Folium Map:

- **Green** — high trust (≥ 0.7)
- **Orange** — medium trust (0.4–0.7)
- **Red** — low trust (< 0.4)
- **Purple grid** — medical deserts

MarkerCluster for scalability to 10 000+ pins.

Stage 5 — Natural Language Query & Dashboard

Query Agent (Groq Llama 3.3 70B):

- Parses natural language → structured filters
- Extracts required capabilities (`has_icu`, `has_dialysis...`)
- Ranks by trust score + distance
- Returns evidence sentences per result

"ICU hospital in Maharashtra" → 7 ranked results

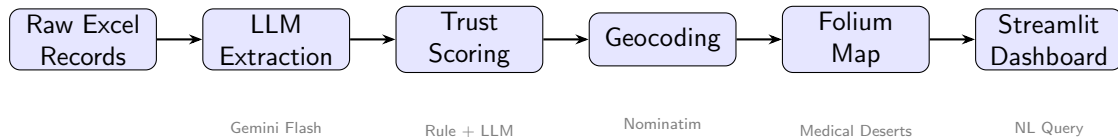
Streamlit Dashboard — 3 tabs:

- 1 **Map** — full Folium embed
- 2 **Query** — NL search interface
- 3 **Data Explorer** — filters, charts, table

Tech stack:

Gemini · Groq · Pandas · Pydantic
Geopy · Folium · Streamlit · Python 3.9+

Pipeline Overview



- Single command: `python run_all.py`
- No database, no manual labels — raw text in, trusted geodata out
- 73 facilities processed end-to-end on free API tier; built for 10 000+